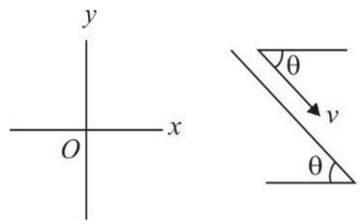
Solutions to JEE Advanced – 1 | JEE 2022 | Paper-1

PHYSICS

Single Correct (6 Questions)

1.(B) $\vec{v}_m = v\cos\theta\,\hat{i} - v\sin\theta\,\hat{j} = \frac{6}{2}\hat{i} - \frac{6\sqrt{3}}{2}\hat{j} = 3\hat{i} - 3\sqrt{3}\hat{j}$ (velocity of man)



$\vec{v}_R = -v'\hat{j}$ (velocity of rain)

$\vec{v}_{Rm} = \vec{v}_R - \vec{v}_m = (-v' + 3\sqrt{3})\hat{j} - 3\hat{i}$ (velocity of rain w.r.t. man)

Since $\vec{v}_{Rm}$ is horizontal, $v_y = 0 = -v' + 3\sqrt{3}$

$\Rightarrow \quad v' = 3\sqrt{3}\ \text{ms}^{-1}$ hence $|\vec{v}_R| = 3\sqrt{3}\ \text{ms}^{-1}$

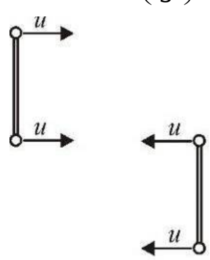
2.(C) Time taken by stone A to reach its highest point, $T = \frac{v_0}{g}$

Since the distance travelled by stone B during time T is h,

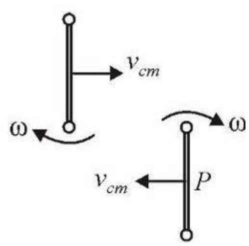
$$v_0 T + \frac{1}{2} g T^2 = h$$

$$\Rightarrow \quad v_0\left(\frac{v_0}{g}\right) + \frac{1}{2} g \left(\frac{v_0}{g}\right)^2 = h \quad \Rightarrow \quad v_0 = \sqrt{\frac{2gh}{3}}$$

3.(C)



Before collision



After collision (due to symmetry)

$$V_{sep} = e\,V_{app}$$

$$\left(\frac{\omega l}{2} - V_{cm}\right) \times 2 = 2u$$

$$\Rightarrow \quad \omega l - 2V_{cm} = 2u \qquad \ldots \text{(i)}$$

Conserving angular momentum about the centre of mass of the rod B (taking clockwise positive)

$$\left(mu\frac{l}{2} + mu \times \frac{3l}{2}\right) + \left(mu \times \frac{l}{2} - mu\frac{l}{2}\right) = \left(\frac{ml^2}{4}\right) \times 2 \times \omega + \left[\left(2 \times \frac{ml^2}{4} \times \omega\right) + 2m \times V_{cm} \times l\right]$$

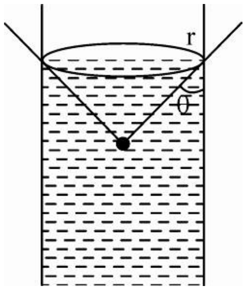
$$\Rightarrow \quad 2mul = ml^2\omega + 2mV_{cm}l \qquad \Rightarrow \quad 2u = \omega l + 2V_{cm} \qquad \ldots \text{(ii)}$$

From (i) and (ii), we get $\omega = \frac{2u}{l}$

4.(B) $2\pi r\cos(\cos^{-1}3/4) = (\pi r^2 \times 6)\rho g$ .....(i)

$2\pi r\cos\theta = (\pi r^2 \times 4)\rho g$ .....(ii)

Solving (i) and (ii) we get $\theta = 60°$



5.(A) At maximum elongation (x_{max}) of spring, speed of m is zero and it moves down by $2x_{max}$

Applying energy conservation: $-mg\,2x_{max} - \frac{1}{2}K[(x_{max})^2 - 0^2] = 0 \Rightarrow x_{max} = \frac{4mg}{k}$

6.(C) Acceleration due to gravity varies with depth d as $g' = g\left(1 - \frac{d}{R}\right)$

As the particle falls into the well, d increases so g' decreases

Multiple correct (6 Questions)

7.(BD) Equation of the wall:

$$y = 20 - x \quad \ldots (1)$$

Equation of trajectory of the projectile:

$$y = x\tan 53° - \frac{10\times x^2}{2\times 250}\times(\sec^2 53°)$$

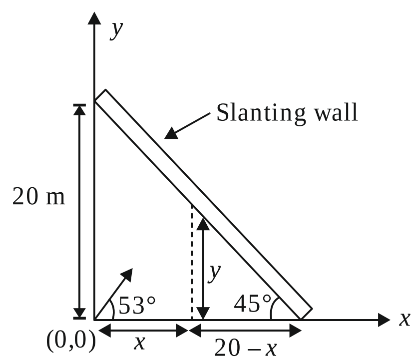
$$y = \frac{4}{3}x - \frac{x^2}{50}\times\frac{25}{9}$$

$$y = \frac{4}{3}x - \frac{x^2}{18}$$

$$20 - x = \frac{4}{3}x - \frac{x^2}{18}$$

$$x^2 - 18x - 24x + 360 = 0$$

$$x^2 - 42x + 360 = 0 \quad \Rightarrow \quad x = 12\text{ m}, 30\text{ m}$$



Valid $x = 12$ m $\Rightarrow$ $y = 8$ m (from (1)) $\therefore$ Coordinates are (12, 8)

Its velocity when it hits the wall

$$u_x = 5\sqrt{10}\cos 53° = 3\sqrt{10}\text{ m/s}$$

$x = 12$ m at time collision $t = \frac{12}{3\sqrt{10}} = \frac{4}{\sqrt{10}}s$

$$u_y = 5\sqrt{10}\times\sin 53° = 4\sqrt{10}\text{ m/s}$$

$$v_y = 4\sqrt{10} - 10\times\frac{4}{\sqrt{10}} = 0;$$

Therefore, the velocity of the projectile when it hits the wall, $\vec{v} = 3\sqrt{10}\,\hat{i}$ m/s

8.(BD) $U = \frac{kr^2}{2} \Rightarrow F = -\frac{dU}{dr} = -kr$; $kr = \frac{mv^2}{r} \Rightarrow mv^2 = kr^2 \Rightarrow v = \sqrt{\frac{k}{m}}r$

$KE = \frac{1}{2}mv^2 = \frac{kr^2}{2}$ $\quad \therefore \quad$ $TE = PE + KE = kr^2$

Time period $= \frac{2\pi r}{v} = 2\pi\sqrt{\frac{m}{k}}$

Angular momentum $= mvr = \sqrt{km}r^2$

9.(AC) Terminal velocity is achieved when net force becomes zero

$\Rightarrow \quad 6\pi\eta rv + \frac{d}{\rho}mg = mg \quad \Rightarrow \quad v = \frac{2(\rho - d)r^2 g}{9\eta} \quad \Rightarrow \quad v = 9\,\text{cm/s}$

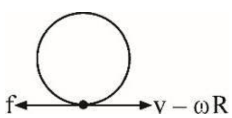
Power after attaining terminal velocity

$\Rightarrow \quad P = \vec{F}_d \cdot \vec{v}_T = -F_d v_T \quad \Rightarrow \quad F_d = 6\pi\frac{2}{3}\times\frac{2}{1000}\times 0.09 = \frac{24\pi}{3\times 1000}\times 9\times 10^{-2}$

$\Rightarrow \quad P = -(72\pi\times 10^{-5})(0.09) = -72\times\frac{25}{8}\times 9\times 10^{-7} = -2.025\times 10^{-4}$

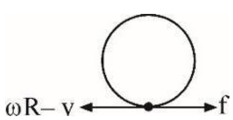
10.(ABD) If $V = \omega R$, there is no slipping and hence displacement of point of application of friction is zero at all moments $\therefore \quad W_f = 0$

If $V > \omega R$, there will be forward slipping and friction acts backwards



$\therefore \quad W_f = -ve$

If $V < \omega R$, there will be backward slipping & friction acts forwards

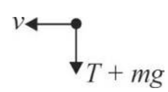


$\therefore \quad W_f = -ve$

(Note that $dw = \vec{F}\,.\,\vec{ds}$, where $\vec{dS}$ is the displacement of point of application of $\vec{F}$ (and not of centre of mass). Since surface is fixed, so $w_{f\,(\text{surface})} = 0$

11.(BCD) At C : $T + mg = \frac{mv^2}{\ell}$

$\Rightarrow \quad mg + mg = \frac{mv^2}{\ell} \Rightarrow v = \sqrt{2g\ell}$



Applying energy conservation from A to C

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mg(2\ell)$$

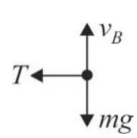
$\Rightarrow \quad u^2 = v^2 + 4g\ell = 2g\ell + 4g\ell \quad \therefore \; u = \sqrt{6g\ell}$

Velocity at B can be found by applying energy conservation from A to B

$\frac{1}{2}mu^2 = \frac{1}{2}mv_B^2 + mg\ell$

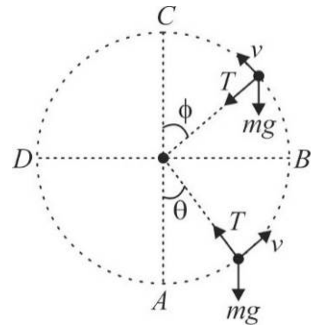
$v_B^2 = u^2 - 2g\ell = 6g\ell - 2g\ell$

$\therefore \quad v_B = 2\sqrt{g\ell}$



Tension provides the required centripetal acceleration

$$\therefore \qquad T = \frac{mv_B^2}{\ell} = \frac{m(4g\ell)}{\ell} = 4mg$$



As bob moves up, its potential energy increases and kinetic energy decreases.
So, speed decreases continuously while going from A to B to C

$$A - B: \; T - mg\cos\theta = \frac{mv^2}{\ell}$$

$$T = mg\cos\theta + \frac{mv^2}{\ell}$$

As θ increases, $\cos\theta$ decreases. As both terms on RHS decreases.
So T decrease

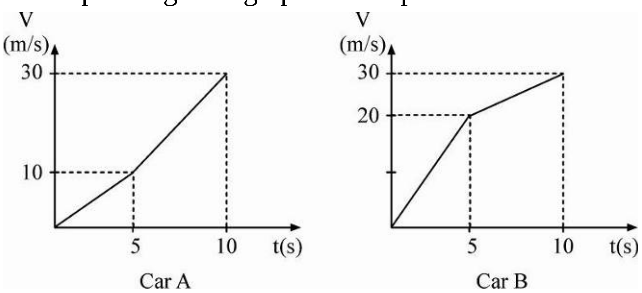
$$B - C: \; mg\cos\phi + T = \frac{mv^2}{\ell}$$

$$T = \frac{mv^2}{\ell} - mg\cos\phi$$

As ϕ decreases, $\cos\phi$ increases.

As first term in RHS decreases and second term increases. So T decreases

12.(AC) Since area under $a-t$graph is same for both, so change in velocity is same.
Corresponding $v-t$ graph can be plotted as



Since area under $v-t$ graph is more for car B, so it travels more distance

Numerical

13.(1.50) Using the parallelogram law of vector addition,

$$R_1 = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 60^\circ} = \sqrt{F_1^2 + F_2^2 + F_1F_2}$$

$$R_2 = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 120^\circ} = \sqrt{F_1^2 + F_2^2 - F_1F_2}$$

Therefore, $\dfrac{F_1^2 + F_2^2 + F_1F_2}{F_1^2 + F_2^2 - F_1F_2} = \left(\dfrac{R_1}{R_2}\right)^2 = \dfrac{19}{7} \qquad \Rightarrow \qquad \dfrac{\left(\dfrac{F_1}{F_2}\right)^2 + 1 + \dfrac{F_1}{F_2}}{\left(\dfrac{F_1}{F_2}\right)^2 + 1 - \dfrac{F_1}{F_2}} = \dfrac{19}{7}$

Now, let $\dfrac{F_1}{F_2} = K$

Therefore, $\dfrac{K^2 + K + 1}{K^2 - K + 1} = \dfrac{19}{7} \qquad \Rightarrow \qquad 12K^2 - 26K + 12 = 0$

Solving, we get $K = \dfrac{3}{2}$ or $\dfrac{2}{3} \qquad \therefore \qquad F_1 > F_2 \Rightarrow K > 1 \;\; \therefore K = \dfrac{3}{2} = 1.5$

14.(7.33) Let the acceleration of the ball be a upwards. Then, the acceleration of the rod will be $2a$ downwards.
So, w.r.t rod, ball moves up with an acceleration $3a$

$$\therefore \quad t = \sqrt{\frac{2L}{3a}}$$

Let, mass of rod = m, then, mass of ball = $\frac{3}{2}m$

$$T - \frac{3}{2}mg = \frac{3}{2}ma \qquad \ldots\text{(i)}$$

$$mg - \frac{T}{2} = 2ma \quad \Rightarrow \quad 2mg - T = 4ma \qquad \ldots\text{(ii)}$$

Adding (i) and (ii), we get

$$\frac{mg}{2} = \frac{11}{2}ma \quad \Rightarrow \quad a = \frac{g}{11}$$

Therefore, $t = \sqrt{\dfrac{2L}{3\left(\dfrac{g}{11}\right)}} = \sqrt{\dfrac{22L}{3g}}$

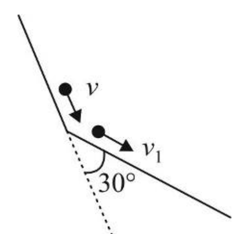
15.(400) $mg = \rho_\omega(0.2v)g$; $\quad mg = \rho_0(0.5v)g$

$$\therefore \quad 0.5\rho_0 = 0.2\rho_\omega \quad \Rightarrow \quad \rho_0 = \frac{2}{5}\times 1000 = 400 \text{ kg/m}^3$$

16.(15) Speed just before reaching B is given by energy conservation

$$mg(5) = \frac{1}{2}mv^2 \quad \Rightarrow \quad v = \sqrt{2(10)(5)} = 10 \text{ m/s}$$

After collision at B, velocity perpendicular to the incline becomes zero while velocity along the incline remains unchanged



$$\therefore \quad v_1 = v\cos 30° = 5\sqrt{3} \text{ m/s}$$

Velocity upon reaching C can be found by applying energy conservation again.

$$mg(7.5) = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\Rightarrow \quad v_2^2 = v_1^2 + 2g(7.5) = 75 + 150 = 225$$

$$\therefore \quad v_2 = 15 \text{ m/s}$$

17.(1.40) For no sliding between B and A $m_B\omega^2 r \le \mu_{AB} m_B g$

$$\Rightarrow \quad \omega_{\max} = \sqrt{(0.2)(9.8)} = 1.4 \text{ rad/s}$$

For no sliding between A and table $(m_A + m_B)\omega^2 r \le \mu_{AT}(m_A + m_B)g$

$$\Rightarrow \quad \omega_{\max} = \sqrt{(0.45)(9.8)} = 2.1 \text{ rad/s}$$

$\therefore$ Maximum ω for no sliding at both contacts is 1.4 rad/s

18.(0.50) From conservation of angular momentum.

$$\frac{M}{2}\sqrt{2gR}R = \frac{MR^2}{2}\omega + \frac{M}{2}(R\omega)R \quad \Rightarrow \quad \omega = \sqrt{\frac{g}{2R}} \Rightarrow K = 0.50$$

CHEMISTRY

1.(A) Let empirical formula of the compound is $C_xH_yO_z$ and one volume of it is used

$$CxHyO_z + (x + y/4 - z/2)O_2 \rightarrow xCO_2 + y/2\,H_2O$$

Volume of O_2 required is, (x + y/4 – z/2) = 2.5 …………………...... (1)

Volume of CO_2 formed is, x = 2 .... (2)

Volume of water vapor formed is, y/2 = 2 .... (3)

x = 2, y = 4 and z = 1

2.(C) Sn(solid, grey) → Sn(solid, white)

$\Delta H = \Delta U + P\Delta V$

$\Delta U = \Delta H - P\Delta V \cong \Delta H$, Because insignificant change in volume.

3.(A) According to VSEPR theory electronegativity of surrounding atom also affect B-A-B bond angle in AB_n type species. F is more electronegative than N hence repulsion between bond pairs is less in NF_3 than in NH_3.

4.(B) Group 17 contains examples of elements that are gas(F_2), liquid (Br_2) and solid(I_2).

5.(C) Because of pπ-dπ back bonding between 2p of N and 3d of Si, $N(SiMe_3)_3$ has trigonal planar geometry around nitrogen and do not acts as Lewis base due to absence of lone pair of electron.

6.(A) $\frac{1}{300} = \frac{1}{760} + \frac{1}{x}$

$\frac{1}{300} - \frac{1}{760} = \frac{1}{x}$

$\frac{760 - 300}{760 \times 300} = \frac{460}{760 \times 300} = \frac{1}{x}, x = 495.65 \cong 496\text{nm}$

7.(ABCD)

Paramagnetic substances are attracted by magnetic field while diamagnetic substances are repelled by magnetic field. H_2O and Zn are diamagnetic and O_2 & NO_2 are paramagnetic substances.

8.(ABCD)

For reversible isothermal expansion T is constant hence, PV = Constant,

During isothermal expansion entropy increases.

w = –[n R T ln V_2 – n R T ln V_1]

9.(AB) HBr is less polar than HF

Order of dipole moment is $HF > H_2O > H_2S$

CuCl is more covalent than NaCl

10.(ABC)

F is most electronegative elements and Cl has maximum value of electron affinity. Cl can expand octet due to the presence of vacant d orbitals. Both HOF and HOCl are acidic compounds.

11.(ABD)

Vinyl group-CH_2=CH-,

Isopropyl group-$(CH_3)_2CH$-

There is only one allylic H atom

IUPAC name is 3-Methyl-But-1-ene

12.(ABC)

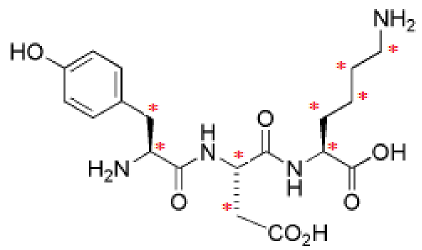
	Fe	Fe^{2+}
Electronic configuration	$[Ar]\ 3d^6\ 4s^2$	$[Ar]\ 3d^6$
Number of unpaired electrons	4	4
Spin only magnetic moment	$\sqrt{24}$ BM	$\sqrt{24}$ BM
Size	larger	smaller

13.(37) $Z_1 = 10$, $Z_2 = 9$, $Z_3 = 18$

C of C = O and benzene ring are sp^2

Two lone pair on every O and one lone pair on every N

* sp^3 carbon atom



14.(3061.83)

$H_2(g) \rightarrow 2H(g)$: E = 435.8 kJ

$2H(g) \rightarrow 2H^+(g) + 2e^-$: $E = 2 \times (2.18 \times 10^{-18}) \times 6.023 \times 10^{23} \times 10^{-3}$ kJ = 2626.028 kJ

Total energy required = 435.8 + 2626.028 = 3061.828 = 3061.83 kJ

15.(935) $\Delta_r H^0 = -394 - (-581) = 187$ kJ

$\Delta_r S^0 = (210 + 52) - (56 + 6) = 200$ J = 0.2 kJ

The reduction of cassiterite by coke is spontaneous at T greater than T_{switch}

$T_{switch} = \Delta_r H^0 / \Delta_r S^0 = 935$ K

16.(93) Mol of P used = Mol of Q formed

Mass of P used = Volume × density = 93 × 1 = 93 g

Mol of P used = 93/ 93 = 1.0 mol

Mol of Q formed = 1.0 (assuming 100 % yield)

Mol of Q formed = 1.0 × 0.75 = 0.75 (considering 75 % yield)

Mol of Q used = Mol of R formed = 0.75 (assuming 100 % yield)

Mol of R formed = 0.75 × 0.5 = 0.375 (assuming 50 % yield)

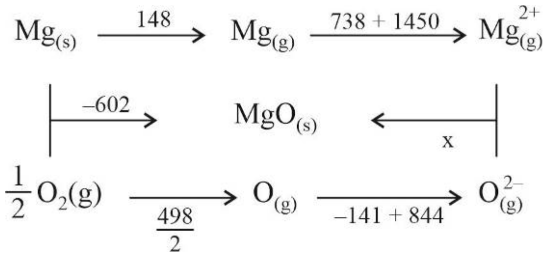
Mass of R formed = Mol of R formed × Molar mass = 0.375 × 248 = 93 g

17.(0.20) $Kc = \dfrac{[H^+]^2}{[Zn^{2+}][H_2S]}$

$[H^+]^2 = Kc \times [Zn^{2+}] \times [H_2S] = 0.04$

$[H^+] = 0.2M$

18.(3890)



$$-602 = 148 + 738 + 1450 - x + \frac{498}{2} - 141 + 844$$

$$x = 3890\,\text{kJ / mol}$$

MATHEMATICS

1.(C) End points of latus rectum given by

$$\left(\pm ae, \pm\frac{b^2}{a}\right) a > b$$

$$h = \pm\sqrt{a^2 - b^2},\ k = \pm\frac{b^2}{a}$$

Let $b^2 = 1 - a^2$

$$h^2 = 2a^2 - 1,\ k = \pm\left(\frac{1-a^2}{a}\right) \Rightarrow k^2 = \frac{(1-a^2)^2}{a^2}$$

$$\left(\frac{1+h^2}{2}\right)k^2 = \left(1 - \frac{1+h^2}{2}\right)^2 = \left(\frac{1-h^2}{2}\right)^2$$

$$2(1+x^2)y^2 = (1-x^2)^2$$

2.(B) $3 \le \left|13 - |z_2|\right| = \left||z_1| - |z_2|\right| \le \left|z_1 - (z_2 + 3 - 4i) + 3 - 4i\right|$

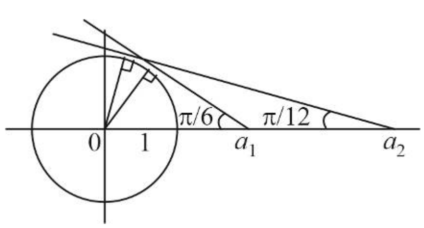
$$\le |z_1| + |z_2 + 3 - 4i| + |3 - 4i| = 13 + 5 + 5 = 23$$

As $\left||z_2| - 5\right| \le \left|z_2 + 3 - 4i\right| = 5 \le |z_2| + |3 - 4i| = |z_2| + 5$

$0 \le |z_2| \le 10$

3.(A) $\omega + \omega^{\left(1-\frac{3}{4}\right)^{-2}} = \omega + \omega^{16} = \omega + \omega = 2\omega$

4.(C)



$$a_1 = \operatorname{cosec}\frac{\pi}{6} = 2,\ a_2 = \operatorname{cosec}\left(\frac{\pi}{12}\right) = \sqrt{6} + \sqrt{2}$$

5.(D) Let common tangent be $y = mx + \frac{2}{m}$ & $\frac{4}{m^2} = 4m^2 + 15$

$\Rightarrow \quad 4m^4 + 15m^2 - 4 = 0$

$\Rightarrow \quad 4m^4 + 16m^2 - m^2 - 4 = 0$

$\Rightarrow \quad (4m^2 - 1)(m^2 + 4) = 0$

$m^2 = \frac{1}{4} \Rightarrow m = \pm\frac{1}{2}$

$c = \pm 4$

Points of contact of $y = \frac{1}{2}x + 4$ given by

$$P\left(\frac{2}{(1/2)^2}, \frac{2\cdot 2}{1/2}\right) = (8, 8) \quad \& \quad Q\left(\frac{-\frac{1}{2}\cdot 4}{4}, \frac{15}{4}\right) = \left(-\frac{1}{2}, \frac{15}{4}\right)$$

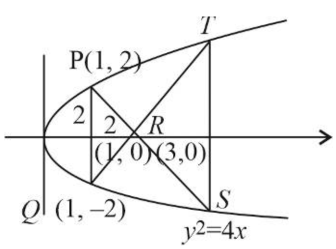
Distance between them $PQ = \sqrt{\left(8+\frac{1}{2}\right)^2 + \left(8 - \frac{15}{4}\right)^2} = \sqrt{\frac{289}{4} + \frac{289}{16}} = \frac{17}{4}\sqrt{5}$

6.(C) $P(h,k)$, chord of contact w.r.t $y^2 = 4ax$ is $ky = 2a(x+h)$ i.e.

$x = \frac{k}{2a}y - h$ touches $x^2 = 4by$ if

$-h = \frac{b}{\frac{k}{2a}} \Rightarrow hk = -2ab$ i.e. locus is $ky = -2ab$

7.(AB)



$P(t=1) \Rightarrow S\left(t = -1 - \frac{2}{1} = -3\right) \equiv (9, -6)$

$\Rightarrow \quad T(9,\ 6)$

$arPQST = \frac{1}{2}\times(4+12)\times 8 = 64\ sq.units$

Equation of circumcircle of $PQST$ is

$$(x - 2y + 3)(x + 2y + 3) + \lambda(x-1)(x-9) = 0$$

$\lambda + 1 = -4 \Rightarrow \lambda = -5$

$x^2 - 4y^2 + 6x + 9 - 5(x^2 - 10x + 9) = 0$

$x^2 + y^2 - 14x + 9 = 0$

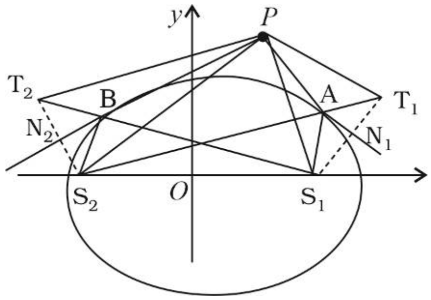
8.(CD) Let $\alpha \in R$ be real root

Then $\alpha^2 - 3\alpha + m = 0$ & $-\alpha + 2 = 0$

$\alpha = 2,\ m = 2$

Second root $= 1 + i = \beta$

9.(ABCD)



$\Delta PS_1N_1 \cong \Delta PT_1N_1$ (as T_1 is image of S_1 about PA)

$\Delta PS_2N_2 \cong \Delta PT_2N_2$ (as T_2 is image of S_2 about PB)

$PS_2 = PT_2,\ PT_1 = PS_1$

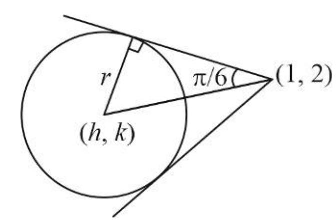
$S_2T_1 = S_2A + AT_1 = S_2A + AS_1 = 2a$

$S_1T_2 = S_1B + BT_2 = S_1B + BS_2 = 2a$

$\Rightarrow \quad \Delta PS_2T_1 \cong \Delta PT_2S_1$

Foot of perpendicular of foci to any tangent lies on auxiliary circle $\Rightarrow$ $N_1,\ N_2$ lies on $x^2 + y^2 = a^2$

10.(ACD) Let centre be $C_1(h, k)$ its distance from $C_2(1, 2)$



$$\sqrt{(h-1)^2 + (k-2)^2} = \frac{\sqrt{27}}{2} - r$$

$$\sqrt{(h-1)^2 + (k-2)^2} = 2r$$

$$\Rightarrow \quad (h-1)^2 + (k-2)^2 = 3$$

Locus $(x-1)^2 + (y-2)^2 = 3$

It's a director circle of $\dfrac{(x-1)^2}{a^2} + \dfrac{(y-2)^2}{b^2} = 1$

Such that $a^2 + b^2 = 3$

11.(ABD) $\dfrac{z-4}{z-2i}$ is purely imaginary

$$\Rightarrow \quad \frac{w-2}{w-i} = ki,\ k \in R - \{0\}$$

$$\frac{w-2}{w-i} + \frac{\overline{w}-2}{\overline{w}+i} = 0$$

$$w\overline{w} + iw - 2\overline{w} - 2i + w\overline{w} - i\overline{w} - 2w + 2i = 0$$

$w = x + iy$

$$2(x^2 + y^2) - 2y - 2.2x = 0$$

$$x^2 + y^2 - 2x - y = 0$$

If $\dfrac{w}{w-2-i}+\dfrac{\overline{w}}{\overline{w}-2+i}=0$

$w\overline{w}-2w+iw+w\overline{w}-2\overline{w}-i\overline{w}=0$

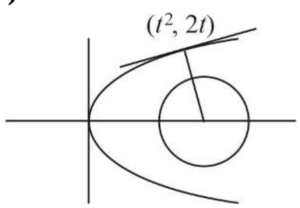
$2w\overline{w}+i(w-\overline{w})-2(w+\overline{w})=0$

12.(AC) $x+y=2(t^2+1),\ x-y=2t$

$\dfrac{x+y}{2}-1=\left(\dfrac{x-y}{2}\right)^2 \Rightarrow (x-y)^2=2(x+y-2)$

$\Rightarrow \left(\dfrac{x-y}{\sqrt{2}}\right)^2=\sqrt{2}\left(\dfrac{x+y-2}{\sqrt{2}}\right)$

13.(2.24)



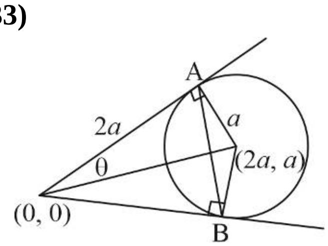
$tx+y=t^3+2t$

$t^3-4t=0$

$t=0,\ 2,\ -2$

Minimum distance $=\sqrt{20}-\sqrt{5}=\sqrt{5}$

14.(1.33)



$AB=2\times 2a\sin\theta=4a\dfrac{a}{\sqrt{5}a}=\dfrac{4a}{\sqrt{5}}$

15.(105)

$z_1\overline{z}_1=1,\ z_2\overline{z}_2=3,\ z_3\overline{z}_3=5,\ z_4\overline{z}_4=7$

$|105z_1+35z_2+21z_3+15z_4|$

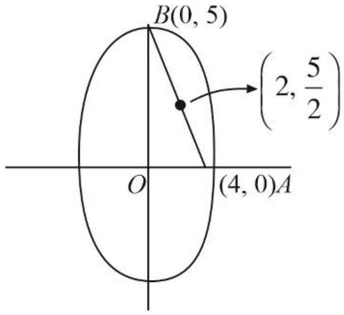
$=\left|\dfrac{105}{\overline{z}_1}+35\cdot\dfrac{3}{\overline{z}_2}+21\cdot\dfrac{5}{\overline{z}_3}+15\cdot\dfrac{7}{\overline{z}_4}\right|$

$=105\left|\dfrac{1}{\overline{z}_1}+\dfrac{1}{\overline{z}_2}+\dfrac{1}{\overline{z}_3}+\dfrac{1}{\overline{z}_4}\right|$

$=105\dfrac{|z_2z_3z_4+z_1z_3z_4+z_1z_2z_4+z_1z_2z_3|}{|z_1||z_2||z_3||z_4|}$

$=\sqrt{105}\,|z_2z_3z_4+z_1z_3z_4+z_1z_2z_4+z_1z_2z_3|$

16.(6.40)



$AB = \sqrt{16+25} = \sqrt{41}$

17.(6) Chord of contact of $(\alpha, 3-\alpha)$ w.r.t $x^2 + y^2 = 9$ is

$$\alpha x + (3-\alpha)y = 9$$

$a(x-y) + 3(y-3) = 0$

$\Rightarrow$ fixed point is (3, 3)

18.(12) $y = mx - am^3 - 2am$ is equation of normal

$\Rightarrow \quad y = 2x - 1\cdot 2^3 - 2\cdot 1\cdot 2 = 2x - 12$